

API Response Time Optimization

1. KPI METADATA SPECIFICATIONS

DATABASE RECORD ID	6a16afa41f233bdd0f87ae96	TARGET CONSTRAINT	200 ms
CURRENT METRIC VALUE	158 ms (Completion: 63%)	PRIORITY CLASSIFICATION	HIGH
SYSTEM PIPELINE STATE	In-Progress	TARGET SLA MILESTONE	2026-06-30
ASSIGNED OPERATOR	John Doe (ID: 6a154bab4f0f9676ea417654)	OBSERVATION TIMESTAMP	2026-05-29 10:15:00 UTC

Registered Technical Objective Statement:

"Optimize Node.js backend endpoints to achieve sub-200ms response times across national data aggregation nodes."

Active Engineer Logs & Status Comments:

"Main routes optimised, auth routes still slow."

2. SYSTEM VALIDATION, DIAGNOSTICS & TELEMETRY DATA

1. Architectural Performance Profiling

Profiling data was compiled inside the staging integration tier for the PrettyPie platform gateway. Analysis isolated processing constraints inside the middleware validation layer and single-threaded payload serialization chains.

FIGURE 1.1: MICROSECOND ALLOCATION TAXONOMY ACROSS CENTRAL ROUTER NODES



2. Applied Engineering Interventions

- Database Layer Calibration:** Appended explicit compound indexing constraints { tenantId: 1, routeHash: 1 } using Mongoose drivers against the main data cache to circumvent heavy collection scanning sequences.

- **Volatile Cache Offloading:** Provisioned a local Redis caching layer configuration running an explicit 300-second eviction TTL sequence for localized government reference lookups.
- **Cryptographic Bottleneck Analysis:** Traced the `auth` layer latency to asynchronous blockages during high-iteration password hashing runs; optimization planning has been moved to a separate sub-task queue.

3. Production Run Benchmark Performance

TRANSACTION UNIFORM RESOURCE IDENTIFIER (URI)	BASELINE PROFILE	TARGET SLA BOUNDARY	POST-FIX ACTUAL
GET /api/v1/services/malaysia-unified/status	412.50 ms	< 200.00 ms	158.00 ms [Within Bounds]
POST /api/v1/auth/gateway-token/exchange	310.20 ms	< 200.00 ms	245.00 ms [Degraded State]

4. Diagnostic Telemetry Stream Output

```
[2026-05-29T10:11:42.112Z] INITIATING TELEMETRY CAPTURE - STREAM_SAMPLES=10000 CONCURRENT CONNS
[2026-05-29T10:12:01.002Z] upstream_time=0.158s backend=prettypie-core status=200 route="GET /
api/v1/services"
[2026-05-29T10:12:05.415Z] upstream_time=0.245s backend=prettypie-auth status=200 route="POST /
api/v1/auth"
[2026-05-29T10:15:00.000Z] MEDIAN AGGREGATION ENGINE: Verified at 158.00ms running mean
profile. Task state mapped to 63% completion pool.
```